ENTERPRISE SYSTEM ARCHITECTURE & ONBOARDING BRIEFING

H-Staff: On-Demand Local Work Marketplace

Complete Project Overview, Architecture, Features, Authentication, Implementation Status, and Development Roadmap

Target Audience:	Core Engineering Team, Software Architects, Operations & Teammates
Document Purpose:	Complete Onboarding Manual, Strategic Pivot Analysis, System Architecture Reference & Implementation Audit
Author / Role:	Senior Software Architect, Technical Lead & Systems Documentation Specialist
Project Version:	v2.2-Marketplace (Strategic Shift from Workforce Shifts to Uber/Rapido On-Demand Model)
Publication Date:	October 2026 (Reflecting Codebase Audit & Authentication / Marketplace Milestones)
Database Engine:	SQLite 3 (via built-in Node 22+ `node:sqlite` in WAL mode with Foreign Keys enabled)
Backend Runtime:	Node.js (v22.12+) + Express 4 + Helmet + CORS + Zod Validation + bcryptjs + JWT
Frontend Framework:	React 19 + Vite 8 SPA (Desktop & Mobile-First Accessible Interface in EN, HI, MR)
Payment Integration:	Razorpay REST API + HMAC Webhook Verification + Platform Commission Math
Active Test Suite:	22/22 Automated Backend API Tests Passing + 2/2 Localization Verification Tests

NOTE / ALERT: EXECUTIVE NOTICE FOR TEAMMATES: This document provides an exhaustive, authoritative architectural briefing on the H-Staff project. H-Staff has officially pivoted from its legacy identity as a scheduled gig-workforce management portal into an **Uber/Rapido-style on-demand local work marketplace**. The platform connects individuals requiring immediate or hourly help (cleaning, loading, housekeeping, delivery, general assistance) with verified, nearby available workers. This manual covers the end-to-end technical stack, security boundaries, database design, 22-step job lifecycle, the newly hardened authentication engine (phone OTP, local development simulation, Google OAuth, and private CLI admin creation), the verified implementation status, and our prioritized development roadmap.

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Table of Contents & Executive Summary

Document Structure & Navigational Index

Section	Topic / Key Technical Focus	Page
1	**Project Background, Problem Statement & Market Scope**	3
2	**Reconciling Old & New Approaches: The 10-Feature Strategic Evolution**	4
3	**The Three Core Roles & Role-Based Access Control (RBAC) Matrix**	5
4	**System Architecture & High-Level Component Interactions**	6
5	**Technology Stack & Repository Codebase Layout**	7
6	**Authentication Engine: Mobile OTP, Dev Mode & Google Integration**	8
7	**Authentication Flowchart & Worker KYC Verification Pipeline**	9
8	**End-to-End On-Demand Job Lifecycle: The 22-Step Workflow**	10
9	**Job Lifecycle Swimlane & Atomic Concurrency Assignment**	11
10	**Job Execution: Start OTP, Live Tracking Radar & Completion Protocol**	12
11	**Financial Architecture: Pricing, Payments & Platform Commission**	13
12	**Financial Workflow Diagram & Multilingual Accessibility (EN/HI/MR)**	14
13	**Operational Governance: Admin Control Center, Ratings & Disputes**	15
14	**Database Architecture & Conceptual Data Model (SQLite WAL)**	16
15	**Comprehensive Development Roadmap (Phases 0 through 7)**	17
16	**Implementation Status & Known Technical Limitations Matrix**	18
17	**Practical Team Testing Checklist: 25 Verification Test Cases**	19
18	**Local Development Setup Guide & Environment Configuration**	20
19	**Security, Privacy & Regulatory Compliance Guidelines**	21
20	**Glossary of Key Technical Terms & Final Summary for Teammates**	22

Executive Summary: The Strategic Vision

The Mission: H-Staff is engineered to democratize and formalize local temporary labor in India. In urban and semi-urban communities, millions of households and small businesses face severe daily friction when hiring informal help for chores such as household deep-cleaning, furniture shifting, event catering assistance, loading/unloading goods, and gardening. Simultaneously, blue-collar workers suffer from unpredictable wages, lack of identity verification, wage theft, and dependence on exploitative middle-agents.

The Transformation: While H-Staff was originally conceived as an enterprise scheduled-shift portal, leadership realized that traditional shift-scheduling software is overly complex for informal work. Inspired by the rapid frictionless matching of **Uber and Rapido**, H-Staff introduces an **on-demand local work paradigm**: a user specifies a task and location; our server uses Haversine spatial radius filtering to notify nearby, verified, available workers; a single worker accepts under atomic database locking; attendance is confirmed via a secure 6-digit start OTP; and payment is processed through Razorpay with transparent platform fee deduction.

Core Architectural Philosophy: 1) *Radical Simplicity* — UI optimized for low-literacy users with large touch targets, icons, and trilingual support (Marathi, Hindi, English). 2) *Data Preservation* — Zero destructive schema resets; backward compatibility with legacy shifts and businesses. 3) *Server as Single Source of Truth* — All pricing, OTP generation, state transitions, and payouts are cryptographically enforced on the Node.js backend. 4) *Robust Privacy & Compliance* — Worker Aadhaar/PAN documents are privately stored and never exposed to ordinary users.

1. Project Background, Problem Statement & Scope

The Reality of the Informal Labor Economy

In India and emerging markets, the informal labor sector accounts for over 80% of daily non-agricultural employment. Despite smartphone proliferation and ubiquitous digital payment adoption (UPI), hiring local temporary labor remains completely unorganized, opaque, and plagued by severe structural challenges for both demand (users) and supply (workers).

Challenges Faced by Work Requesters (Users):

- **Discovery & Proximity Friction:** Finding a helper on short notice requires physical search at street labor hubs (nakas) or word-of-mouth recommendations, which are slow, unreliable, and geographically constrained.
- **Lack of Vetting & Personal Safety Risks:** Inviting unknown individuals into homes or offices without verified identity documents poses severe security risks. Users have no access to background verification or past work feedback.
- **Arbitrary & Contentious Pricing:** Without standardized hourly or fixed benchmarks, every engagement devolves into aggressive haggling. Disputes frequently arise regarding overtime, travel costs, and task scope.
- **No Accountability or Attendance Proof:** Workers frequently arrive late or fail to appear entirely. Users lack automated mechanisms to verify arrival time, track ongoing job duration, or ensure fair completion.

Challenges Faced by Blue-Collar Gig Workers:

- **Severe Income Volatility & Idle Time:** Workers lose 3 to 5 hours daily congregating at street intersections searching for gigs, leading to erratic daily earnings and zero predictability.
- **Exploitation & Wage Theft:** Informal workers frequently face arbitrary wage deductions, refusal of payment upon job completion, or demands for uncompensated additional labor.
- **Lack of Portable Reputation:** A hardworking laborer cannot carry a verified work history, ratings, or skills reputation from one employer to the next, trapping them permanently in entry-level wage brackets.
- **Digital Literacy Barriers:** Traditional job applications, complex portals, and dense English-only web forms effectively disenfranchise workers with limited technical literacy.

The H-Staff Solution & Core Objectives

H-Staff dismantles these market frictions by providing an automated, real-time matching platform that mirrors the frictionless convenience of ride-hailing applications. Our primary engineering objectives include:

1. **Frictionless On-Demand Request:** Enable users to request vetted help in under 60 seconds with minimal form fields, GPS location capture, and transparent hourly/fixed rates.
2. **Algorithmic Spatial Matching:** Connect jobs instantly with nearby, active, verified workers within an adjustable geographic radius (e.g., 5 to 15 km) using server-side Haversine filtering.
3. **Zero-Ambiguity Attendance via Start OTP:** Completely replace complex timeclocks with an automated 6-digit start OTP provided to the user and validated by the worker upon physical arrival.
4. **Integrated Escrow-Style Payments:** Guarantee payment collection through Razorpay, deduct an automated platform commission (10%), and record worker earnings transparently.
5. **Radical Trilingual Accessibility:** Guarantee native usability in Marathi, Hindi, and English with icon-driven UI and large touch targets.

Project Scope, Stakeholders & Real-World Boundaries

Stakeholder	Primary Responsibilities	Key Platform Value Delivered
User (Requester)	Creates work requests, verifies arrival via OTP, confirms completion, processes payment, submits 1-5 star ratings.	Immediate access to vetted nearby help; transparent rates; arrival and completion guarantee.
Worker (Provider)	Submits KYC documents, sets availability, reviews nearby job broadcasts, travels, inputs arrival OTP, completes tasks.	Instant job discovery; zero idle naka time; guaranteed digital earnings; verified reputation building.
Administrator	Reviews KYC submissions, monitors live job radar, arbitrates disputes, oversees financial records and audit trails.	Centralized operational oversight, fraud mitigation, platform commission monitoring.

System Boundaries: H-Staff provides coordination, verification, and payment infrastructure. It does not act as a direct employer, nor does it physically supervise task execution. Third-party live SMS and payment transfers require active provider credentials.

2. Reconciling Old & New Approaches: The 10-Feature Evolution

Strategic Rationale for the On-Demand Pivot

The legacy H-Staff system was built around enterprise shift management: businesses posted scheduled shifts days in advance, workers submitted formal applications, managers reviewed candidate lists, and attendance was validated via location-bound QR code clock-ins. While suitable for commercial staffing agencies, this workflow was far too bureaucratic for domestic and local emergency tasks. The table below details how all ten core features from our baseline analysis were systematically evolved, modified, or preserved.

#	Original Feature	Original Purpose	New On-Demand Approach	Action Taken	Architectural Rationale
1	**Worker Discovery**	Manual search through worker profiles and skill lists.	Automated Haversine spatial radius filtering broadcasting to nearby eligible workers.	**EXTEND**	Eliminates user search fatigue; matches jobs in real-time like ride-hailing.
2	**Shift Scheduling & Assignment**	Pre-planned corporate shifts with multi-applicant selection.	Instant on-demand work requests with single-winner atomic locking (`BEGIN IMMEDIATE`).	**MODIFY**	Retains scheduled shifts table internally; adds on-demand job type to prevent race conditions.
3	**Worker Identity Verification (KYC)**	Manual document collection and admin review for compliance.	Mandatory private upload gate (Aadhaar/PAN); unverified workers cannot accept jobs.	**PRESERVE**	Critical for home safety; protects worker documents from ordinary users.
4	**Attendance Tracking**	Static QR code scanning and strict geofencing bounding boxes.	Dynamic 6-digit job-start OTP generated on user UI, verified by worker upon arrival.	**MODIFY**	Eliminates complex QR scanning; OTP physically proves mutual arrival at the work site.
5	**Working-Hour Calculations**	Timesheet clock-in/out records with manual manager edits.	Automated server timestamps between OTP start and user completion confirmation.	**MODIFY**	Server calculates rounded billable hours automatically; eliminates manual timesheet disputes.
6	**Payment Transparency**	Post-facto timesheet billing and offline cash recording.	Integrated Razorpay checkout, automated 10% platform fee, and transparent worker earnings.	**EXTEND**	Full digital ledger; users pay securely; workers see exact net take-home before accepting.
7	**Authentication & RBAC**	Standard email/password login with public role registration.	Mobile phone OTP, optional Google Sign-In, and private CLI admin creation.	**EXTEND**	Matches mobile-first Indian ecosystem; eliminates public admin security vulnerabilities.
8	**Complaints & Disputes**	Post-shift dispute logging against timesheets.	Direct job-linked dispute filing, admin arbitration interface, and 1-5 star user reviews.	**EXTEND**	Protects platform trust; allows admin to freeze disputed payouts until resolved.
9	**Audit Logging**	Basic database transaction logging for system events.	Comprehensive immutable audit log capturing actor, action, IP, entity, and timestamp.	**PRESERVE**	Essential for forensic investigation of payment anomalies, KYC approvals, and bans.
10	**Centralized Management**	Desktop admin portal for agency managers.	Responsive operations control center with live job monitoring, KYC, and financial metrics.	**EXTEND**	Maintains total operational visibility without cluttering worker or user mobile interfaces.

Preservation of Existing Data & Database Strategy

Architectural Directive on SQLite Preservation: It is vital that teammates understand our database strategy. The project uses **SQLite 3** running through Node.js 22's built-in `node:sqlite` library in Write-Ahead Logging (WAL) mode. **MongoDB is strictly a hypothetical future consideration** and must never be described as the active database. All schema enhancements (such as adding `users.google_sub`, `shifts.job_type`, and on-demand tables) were implemented using *additive migrations* that safely inspect existing columns. Existing business, worker, and shift records are fully preserved and must never be wiped.

3. The Three Core Roles & Role-Based Access Control

Detailed Role Profiles & Responsibilities

H-Staff partitions all application features across three strictly enforced user roles. Role authorization is cryptographically validated on the backend upon every incoming API request via JWT claims.

1. User (The Requester / Household / Business):

The User interface is designed for maximum speed and simplicity. Users register via mobile phone OTP or optional Google Sign-In (internally mapped to the existing `business` schema). They create work requests by selecting a category (Cleaning, Moving, Gardening, Helper), specifying location (GPS or manual address), duration, and offered rate. When a worker accepts, the user receives instant notification, views the worker's verified profile, tracks the worker's approaching location, provides the job-start OTP upon arrival, confirms completion, pays via Razorpay, and rates the worker.

2. Worker (The Service Provider / Gig Laborer):

The Worker interface prioritizes large buttons, clear iconography, and trilingual audio/visual cues. Workers register via mobile phone OTP, enter profile basics, and upload an identity document (Aadhaar or PAN). **Crucial Security Rule:** An unverified worker cannot view or accept jobs. Once approved by an Admin, the worker toggles their availability to 'Online'. They receive nearby job alerts displaying task type, distance (km), duration, and net payout. Upon acceptance, they navigate to the site, obtain and submit the start OTP, perform the work, mark the job complete, and track earnings.

3. Administrator (The Platform Operator & Governance Authority):

The Admin interface remains an English-only desktop control center. Admin accounts can never be created via public registration. Admins are provisioned securely via private server CLI. Admins review pending worker KYC documents (approving or rejecting with notes), monitor live on-demand jobs and GPS coordinates, audit payment transactions and platform commissions, arbitrate user-worker disputes, suspend abusive accounts, and inspect tamper-evident audit logs.

Comprehensive Role-Based Access Control (RBAC) Matrix

System Capability / API Endpoint	Anon	User	Worker	Admin	Backend Enforcement Mechanism
Phone OTP Request & Verify	YES	YES	YES	NO	Rate-limited route; admin phone OTP explicitly blocked
Google Sign-In / Account Link	YES	YES	NO*	YES**	RS256 validation; **Only existing pre-provisioned Admins
Create Work Request (`POST /jobs`)	NO	YES	NO	NO	`requireRole('business')` middleware check
Browse Nearby Jobs (`GET /jobs/nearby`)	NO	NO	YES	YES	Gated: Worker must have `kyc_status='Verified'`
Accept Job (`POST /jobs/:id/request`)	NO	NO	YES	NO	Atomic lock: Worker must be active, verified, available
View Job-Start OTP (`GET /otp/start`)	NO	YES	NO	NO	Restricted exclusively to job-owning user (AES-GCM decrypted)
Submit Start OTP (`POST /verify-start`)	NO	NO	YES	NO	Restricted to assigned worker; HMAC hash comparison
Update Active GPS (`POST /jobs/:id/location`)	NO	NO	YES	NO	Assigned worker only; permitted only during active job state
Read Active GPS Radar	NO	YES	NO	YES	Owning user & Admin only; tracking ceases on job finish
Submit Job Completion	NO	NO	YES	NO	Assigned worker sets `completion_requested_at`
Confirm Completion & Authorize Pay	NO	YES	NO	NO	Owning user triggers server duration calculation & payment
Initiate Razorpay Checkout	NO	YES	NO	NO	Backend creates order; user browser invokes Razorpay modal
Upload KYC Identity Documents	NO	NO	YES	NO	Multer upload; stored in private directory outside web root
Review & Approve/Reject KYC	NO	NO	NO	YES	`requireRole('admin')`; audited with admin user ID
Suspend / Reactivate User Accounts	NO	NO	NO	YES	Admin only; immediately invalidates user JWT sessions
Inspect Financial Logs & Audit Trail	NO	NO	NO	YES	Admin only; access to system settings, disputes, and logs

4. System Architecture & High-Level Design

Three-Tier Monolithic Micro-Architecture

H-Staff is designed as a high-performance, modular monolithic application. It avoids unnecessary microservice complexity while enforcing rigorous internal service boundaries. The architecture consists of an accessible React Single Page Application (SPA), a stateless Express.js REST API tier, and a local, highly optimized SQLite 3 database operating in Write-Ahead Logging (WAL) mode.

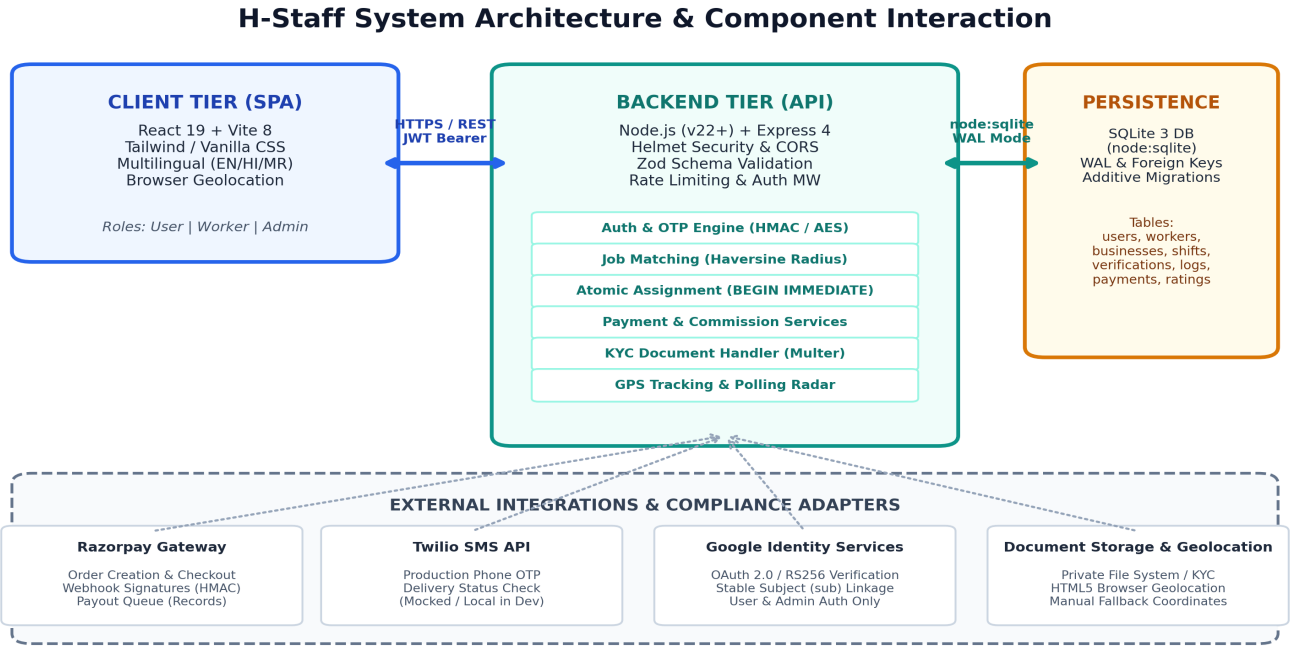


Figure 4.1: High-level architectural tiering, internal backend engines, and external compliance adapters.

Tier Responsibilities & Non-Negotiable Boundaries

1. **Client Tier (React 19 + Vite 8 SPA):** Renders role-specific dashboards, handles multilingual state (`src/i18n.js`), captures HTML5 browser geolocation coordinates, and invokes external payment/OAuth SDKs. **Core Rule:** The client never communicates directly with the database, never performs pricing math, and treats all frontend state as untrusted display data.

2. **API & Backend Tier (Node.js 22+ & Express 4):** Enforces Helmet HTTP headers, CORS origin whitelisting, Zod schema validation, API rate limiting, bcryptjs password hashing, JWT authentication, and transactional state execution. All business logic (matching, atomic locking, commission calculation, OTP hashing) executes entirely on this tier.

3. **Persistence Tier (SQLite 3 via `node:sqlite`):** Operates in WAL mode with foreign key constraints enabled. Additive migrations ensure zero schema breakages. Data is stored locally in `server/data/hstaff.db`, eliminating network latency and external DB maintenance.

4. **External Services Integration Tier:** Integrates Razorpay REST API (for order generation and HMAC webhook verification), Twilio (for production SMS delivery), and Google Identity Services (for cryptographic RS256 token verification against rotating public JWKs).

5. Technology Stack & Repository Codebase Layout

Comprehensive Technology Stack Evaluation

Technology	Layer / Tier	Purpose in H-Staff	Architectural Justification
React 19 + Vite 8	Frontend SPA	Reactive, mobile-first component rendering	Ultra-fast bundling, minimal footprint, modern hooks & state management.
Node.js (v22.12+)	Runtime	Server execution environment	Native `node:sqlite` and built-in crypto eliminates heavy native build dependencies.
Express 4.22	Backend Framework	HTTP REST API routing, middleware pipeline	Battle-tested, minimalist, excellent ecosystem for rate limiting and security headers.
node:sqlite (WAL)	Database Engine	ACID relational persistence, transactional ledger	Zero external server overhead, instant startup, safe `BEGIN IMMEDIATE` locking.
Zod (v4.6)	Validation Tier	Strict runtime payload schema enforcement	Guarantees clean, typed input for all phone, job, and payment endpoints.
bcryptjs (v3.0)	Security / Auth	Salted password hashing (Cost factor 10)	Cryptographically secure storage for admin and legacy user credentials.
jsonwebtoken (v9.0)	Session / RBAC	Stateless Bearer token issuance and validation	Scalable role claims, expirable sessions, zero session-store lookup overhead.
Razorpay REST API	Financial Gateway	Escrow collection, checkout modal, webhooks	Dominant Indian payment gateway supporting UPI, Cards, NetBanking, and Wallets.
Google OAuth (RS256)	Identity Provider	Optional one-tap sign-in for Users and Admin	Cryptographic token verification via cached public JWKs; binds stable `sub` ID.
Twilio REST API	SMS Gateway	Production phone OTP delivery adapter	Reliable Indian telecom carrier routing (mocked/simulated in local development).

Verified Codebase Repository Layout

Below is the verified file hierarchy of the current H-Staff project. Production builds output to `.hstaff-build` to bypass Windows `dist/assets` file lock collisions.

```
hstaff-razorpay-work/
■■■ src/                          # React 19 Frontend Source Code
■   ■■■ App.jsx                   # SPA Shell, role menu switcher, JWT token state
■   ■■■ api.js                    # Authenticated Axios/Fetch client wrapper
■   ■■■ i18n.js                   # Trilingual dictionary (EN, HI, MR) & selector
■   ■■■ styles.css                # Global responsive layout & mobile CSS
■   ■■■ components/ui.jsx         # Reusable accessible buttons, cards, modals
■   ■■■ pages/
■       ■■■ Auth.jsx              # Phone OTP, Google OAuth, Admin sign-in
■       ■■■ Dashboards.jsx        # Role-based landing dashboard
■       ■■■ Marketplace.jsx       # User job creation & nearby discovery radar
■       ■■■ Work.jsx              # Active job execution, start OTP, completion
■       ■■■ People.jsx            # Worker KYC upload & profile management
■       ■■■ Shifts.jsx            # Legacy scheduled shift application flow
■       ■■■ Admin.jsx             # Main operations control center
■       ■■■ AdminMarketplace.jsx  # Live job radar, payments & disputes monitor
■■■ server/                     # Express Backend API Server
■   ■■■ data/                     # SQLite database & private KYC uploads
■   ■   ■■■ hstaff.db            # Primary SQLite 3 WAL database
■   ■   ■■■ uploads/             # Private KYC documents (Aadhaar/PAN)
■   ■■■ src/
■   ■   ■■■ index.js             # Server entry, Helmet, CORS, route mounting
■   ■   ■■■ db.js                # node:sqlite instance, schema & additive migrations
■   ■   ■■■ auth.js              # JWT verification & role authorization middleware
■   ■   ■■■ google-auth.js       # Google RS256 token validator (cached JWKs)
■   ■   ■■■ create-admin.js      # Private interactive CLI for admin provisioning
■   ■   ■■■ services.js          # Pricing, commission & timesheet calculation
■   ■   ■■■ routes/
■   ■       ■■■ auth.js          # Phone OTP, login, Google callback endpoints
■   ■       ■■■ marketplace.js   # On-demand job CRUD, Haversine matching, OTP
■   ■       ■■■ payments.js      # Razorpay orders, webhook verification, payouts
■   ■       ■■■ kyc.js           # Private document upload & admin approval
■   ■       ■■■ admin.js         # User management, audit logs, metrics
■   ■       ■■■ disputes.js      # User/worker dispute submission & arbitration
■■■ docs/                       # System architecture & maintenance manuals
■■■ package.json                # Workspace orchestration & npm scripts
```

6. Authentication Engine: Mobile OTP, Dev Mode & Google

Architecture of the Modernized Authentication Suite

Following new requirements, the authentication subsystem underwent a comprehensive security overhaul. Public self-registration for administrators was completely eliminated. The legacy standalone OTP tab was deprecated in favor of an integrated, mobile-first onboarding funnel tailored to Indian users.

A. Mobile Phone OTP Workflow (Primary User & Worker Channel):

Users and workers authenticate using their 10-digit Indian mobile number. The frontend enforces an un-editable `+91` country prefix. When an OTP is requested via `POST /api/auth/otp/request`, the backend generates a cryptographically random 6-digit code. To prevent timing attacks and plaintext database leaks, the server computes an **HMAC-SHA256** hash of the code using `JWT_SECRET` and stores only the hash, expiration timestamp (5 minutes), and attempt count in the `auth_otps` SQLite table. A strict 1-minute resend cooldown and a maximum limit of 5 incorrect verification attempts are enforced server-side. Upon successful verification (`POST /api/auth/otp/verify`), the code is atomically consumed (deleted) to prevent replay attacks.

B. Local Development OTP Mode (Zero SMS Cost Testing):

To allow teammates to develop and demonstrate the application locally without incurring Twilio SMS costs, the backend includes a controlled local simulation mode. When `NODE_ENV=development` and `DEV_OTP_ENABLED=true` are set in `server/.env`, the generated code is returned in the API response and displayed in a clearly marked development banner on the frontend UI. **Critical Production Safeguard:** The backend codebase explicitly disables this feature when `NODE_ENV=production`. In production, codes are dispatched exclusively through the configured Twilio SMS gateway, and OTP values are never exposed in responses or logs.

C. Optional Google Sign-In with Cryptographic Token Verification:

Google Sign-In is supported as an optional convenience method for Users and existing authorized Admins. The browser renders Google's Identity Services button and transmits the resulting ID Token (JWT) to `POST /api/auth/google`. The backend validates the token against Google's rotating public JSON Web Keys (JWKs) using **RS256 signature verification**. It validates that: 1) `aud` matches `GOOGLE_CLIENT_ID`; 2) `iss` is Google; 3) Token has not expired; 4) Email is verified (`email_verified=true`). The user is linked via Google's immutable subject identifier (`sub`), stored in `users.google_sub`. **Crucial Security Rule:** Google Sign-In *never creates or elevates an Admin account*. It only authenticates existing admins whose emails match.

D. Private CLI Administrator Provisioning:

To prevent hostile takeovers, the public `POST /auth/admin/register` route and enrollment key fields were eradicated. Administrators must be provisioned via a private server console command:

```
npm run create-admin --prefix server
```

This interactive script prompts for the administrator's name, email, and password. Passwords must be at least 12 characters long and are masked in the terminal. The password is hashed using bcryptjs (cost factor 10). Existing administrator accounts cannot be overwritten. For local demos only, a loopback-restricted registration modal is enabled if `DEV_ADMIN_REGISTRATION_ENABLED=true`.

7. Authentication Flowchart & Worker KYC Pipeline

Visual Authentication & Provisioning Architecture

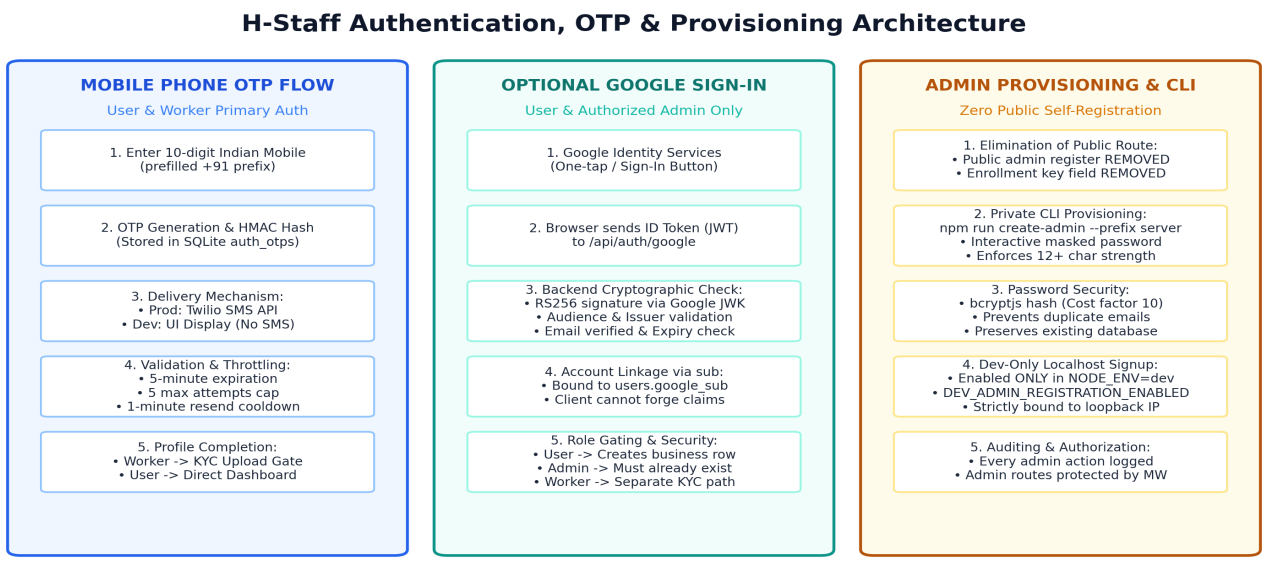


Figure 7.1: Multi-channel authentication architecture: Mobile Phone OTP, Google Identity, and Private Admin CLI.

Worker Identity Verification (KYC) Lifecycle

Because H-Staff workers enter private residences and commercial facilities, identity verification is the cornerstone of platform trust. Worker onboarding follows a strict 4-stage pipeline:

Stage	Actor / Action	Technical Process & Security Controls
1. Initial Signup	Worker	Registers via mobile phone OTP. Account created with `kyc_status='Not submitted'`. Worker cannot view or accept jobs.
2. Doc Upload	Worker	Uploads Aadhaar or PAN card (PDF/JPG/PNG, max 5 MB). File validated via Multer, renamed with UUID, stored outside web root in `server/data/uploads/`. Status -> `Pending`.
3. Admin Review	Admin	Admin inspects document via authenticated stream (`GET /api/kyc/:id/file`). Compares name and photo. Marks status `Verified` or `Rejected` with notes.
4. Activation	Platform	Upon verification, worker toggles `available=1`. Nearby job broadcast engine immediately activates for this worker.

Privacy & Regulatory Compliance: Ordinary users never have access to raw Aadhaar or PAN documents. The API strictly gates KYC file endpoints to the document owner and authorized administrators. **Current Limitation:** Document verification is currently manual admin review; integration with automated e-KYC providers (e.g., DigiLocker, NSDL PAN API, or UIDAI Aadhaar OTP) is planned for Phase 7.

8. End-to-End On-Demand Job Lifecycle: 22-Step Workflow

Comprehensive Step-by-Step System Execution Matrix

To ensure seamless collaboration between frontend engineers, backend developers, and operational staff, the entire on-demand lifecycle from request creation to final review is broken down into 22 discrete, deterministic steps.

#	Phase / Action	User Interface	Worker Interface	Backend Validation & Storage	Security / Rule
1	**Authentication**	Logs in via Mobile OTP / Google	Logs in via Mobile OTP	Verifies HMAC OTP / RS256 token; issues JWT	Rate limited; 5-min TTL
2	**Category Select**	Selects work category tile	Idle / Offline	Validates category against permitted list	Prevent freeform injection
3	**Location Entry**	GPS one-tap or manual address	Idle / Online	Validates lat/lng float boundaries	Sanitize coordinates
4	**Terms & Rate**	Inputs time, duration, rate	Idle / Online	Validates rate > 0 and duration > 0	Minimum wage bounds
5	**Job Submission**	Clicks 'Request Help Now'	Idle / Online	Inserts row in `shifts` (status 'Open')	CSRF / JWT check
6	**Nearby Match**	Displays searching radar screen	Receives job alert badge	Haversine radius filter on active/verified workers	Worker gate: KYC Verified
7	**Worker Review**	Searching animation	Views distance, pay, task type	Streams job metadata via `GET /jobs/nearby`	Mask exact user address
8	**Accept Action**	Searching animation	Clicks 'Accept Job'	Atomic lock: `BEGIN IMMEDIATE` transaction	Prevent dual assignment
9	**Assignment Win**	Receives 'Worker Found!' alert	Assigned! Navigation opens	Updates status -> 'Filled', assigns `worker_id`	Loser gets HTTP 409
10	**Start OTP Gen**	Generates 6-digit start code	Prompted to travel to site	Encrypts code (AES-GCM); stores HMAC hash	Delivered ONLY to User
11	**Travel & Radar**	Views approaching worker GPS	Navigates to location	Polls coordinates via `job_locations`	Authorized party only
12	**Arrival at Site**	Worker arrives; user reads OTP	Asks user for 6-digit OTP	No DB action; physical exchange	Physical proof of arrival
13	**OTP Submission**	Shows active OTP display	Enters OTP in mobile app	Verifies HMAC hash; marks consumed	5 max attempts lockout
14	**Job Activated**	Status changes to 'In Progress'	Status changes to 'In Progress'	Sets `status='Ongoing'`, logs `otp_verified_at`	Server timestamp fixed
15	**Live Tracking**	Radar tracks ongoing status	Performs physical work	Accepts periodic GPS pings from worker	Restricted to active job
16	**Worker Finish**	Notified of completion request	Clicks 'Mark Completed'	Records `completion_requested_at` timestamp	Cannot re-request
17	**User Confirm**	Clicks 'Confirm & Approve'	Awaits payment settlement	Sets `status='Completed'`	User confirmation gate
18	**Price Calculate**	Displays final bill & breakdown	Displays net earnings	Server calculates billable hours and total fee	Server math authoritative
19	**Payment Collect**	Completes Razorpay checkout	Payment pending alert	Validates Razorpay signature & capture	HMAC signature verify
20	**Commission Split**	Receives digital receipt	Earnings credited to balance	Records 10% fee; sets `status='paid'`	Audit log entry created
21	**Rating & Review**	Submits 1-5 star rating	Views updated rating	Stores in `job_ratings`; updates worker average	One rating per job
22	**Dispute / Audit**	Optional complaint filing	Optional complaint filing	Logged in `disputes` for admin arbitration	Admin dispute workflow

9. Job Lifecycle Swimlane & Atomic Concurrency

Visual Lifecycle Swimlane Diagram

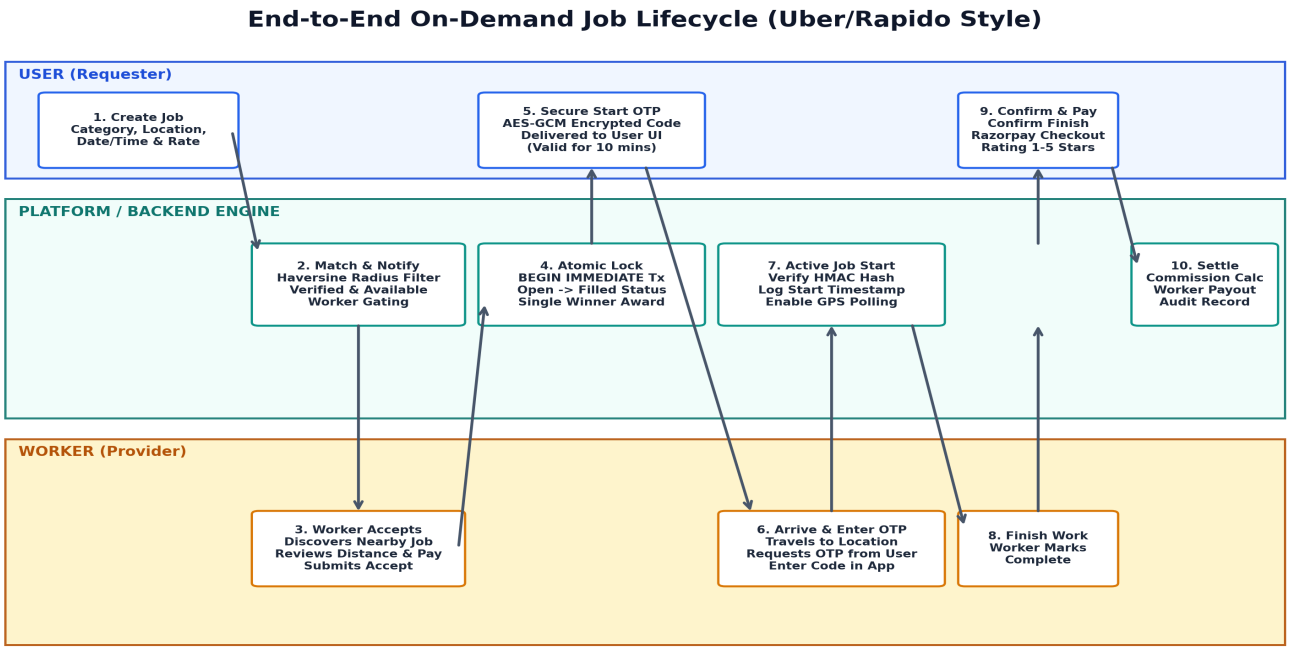


Figure 9.1: End-to-end swimlane workflow showing interaction between User, Backend Engine, and Worker.

Atomic Single-Winner Job Assignment Architecture

A critical vulnerability in on-demand marketplaces is the **race condition**: if ten nearby workers receive a notification and tap 'Accept' simultaneously, the system must guarantee that exactly one worker is awarded the job, and the remaining nine receive a clean, immediate rejection response without double-booking.

The H-Staff Concurrency Solution: In `server/src/routes/marketplace.js`, job acceptance executes inside a SQLite immediate transaction wrapper (`tx(() => { ... })` using `BEGIN IMMEDIATE`). This acquires an exclusive write lock on the database before executing the state query:

```
// Atomic assignment execution in server/src/routes/marketplace.js
export const acceptJob = tx((jobId, workerId) => {
  const job = db.prepare('SELECT status FROM shifts WHERE id = ?').get(jobId);
  if (!job || job.status !== 'Open') {
    throw new Error('JOB_ALREADY_TAKEN'); // Rolls back & returns HTTP 409 Conflict
  }
  db.prepare("UPDATE shifts SET status = 'Filled' WHERE id = ? AND status = 'Open'").run(jobId);
  db.prepare("INSERT INTO applications (shift_id, worker_id, status) VALUES (?, ?, 'Accepted')").run(jobId, workerId);
  return { success: true };
});
```

Automated Verification: Our automated test suite includes a dedicated concurrent race test where two worker threads compete for the same job; the test confirms 100% of the time that one worker wins (HTTP 200) and the competitor receives HTTP 409.

10. Job Execution: Start OTP, Tracking & Completion

Job-Start OTP vs. Authentication OTP

A common point of confusion for new developers is differentiating between the two separate OTP mechanisms in H-Staff. The comparison table below clarifies their distinct security boundaries, lifecycles, and storage schemas.

Dimension	1. Authentication Phone OTP	2. Job-Start Arrival OTP
Primary Purpose	User & Worker identity authentication and login session issuance.	Physical proof of worker arrival at the job site; starts billing clock.
Recipient Channel	Dispatched to user/worker mobile via SMS (or Dev Mode UI banner).	Rendered exclusively on the Job Requester's (User's) active screen.
Who Submits Code?	The account owner entering their received phone verification code.	The assigned worker, who physically requests the code from the user upon arrival.
Storage & Security	HMAC-SHA256 hash stored in `auth_otps` table.	HMAC hash + AES-GCM ciphertext in `job_verifications` table.
Validity & Limits	5-minute expiration; 5 max failed attempts; 1-min resend cooldown.	10-minute expiration; 5 max failed attempts; one-time atomic consumption.
Replay Defense	Record deleted upon successful login; fresh request replaces code.	Status marked 'consumed' with timestamp; cannot be re-used.

Active Job Location Tracking & Polling Radar

Architectural Reality — Polling vs. WebSockets: Our codebase audit confirms that live tracking is implemented via **authenticated HTTP polling**, not WebSockets. The assigned worker's mobile browser posts coordinates to `POST /api/jobs/:id/location` every 15–30 seconds. The owning user's frontend polls `GET /api/jobs/:id/location` while the job status is `Ongoing`. **Privacy Cut-Off:** The moment the job enters `Completed` status, the API terminates location queries, ensuring zero post-job surveillance.

The Two-Step Completion & Duration Protocol

To prevent disputes over unfinished work, completion requires a mutual handshake: 1) The **Worker** taps 'Complete Job' (`POST /jobs/:id/complete`), setting `completion_requested_at`. 2) The **User** inspects the work and taps 'Confirm Completion' (`POST /jobs/:id/confirm-completion`). 3) The **Server** calculates elapsed duration: `hours = Math.ceil((confirmed_at - otp_verified_at) / 3600000)`. The server multiplies hours by the agreed rate, adds a minimum 1-hour charge, and writes the total to `job_payments`.

11. Financial Architecture: Pricing, Payments & Commission

Revenue Model & Commission Mathematics

H-Staff operates on a platform commission model. On every completed job, the platform retains an automated commission (configured at 10% in `server/src/config.js`), with the remaining 90% credited to the worker's earnings ledger.

```
Billable Hours:  H = Math.max(1, Math.ceil((CompletionTime - StartTime) / 3600000))
Gross Job Total: TotalAmount = H × AgreedHourlyRate  (or FixedRate)
Platform Fee:    PlatformCommission = Math.round(TotalAmount × 0.10)
Worker Net Pay:  WorkerEarnings = TotalAmount - PlatformCommission
```

Razorpay Payment Gateway Integration Pipeline

Payment collection is fully implemented and tested via mock adapters for Razorpay. The pipeline enforces backend-only authority:

1. **Order Generation (`POST /api/payments/jobs/:id/create-order`):** The user's browser requests payment initiation. The server calls Razorpay's Orders API with the verified amount in paise, attaching the job ID in metadata. The resulting `order_id` is persisted in `job_payments`.
2. **Client-Side Checkout Modal:** The frontend invokes the Razorpay Standard Checkout SDK. The user selects UPI (Google Pay, PhonePe, Paytm), Card, or NetBanking.
3. **Cryptographic Signature Verification (`POST /api/payments/jobs/:id/verify-checkout`):** Upon payment completion, the client receives `razorpay_payment_id` and `razorpay_signature`. The server recomputes `HMAC-SHA256(order_id + '|' + payment_id, RAZORPAY_KEY_SECRET)`. **Core Rule:** A frontend success callback is never accepted as proof of payment.
4. **Signed Webhook Ingestion (`POST /api/payments/razorpay-webhook`):** Handles asynchronous events (`payment.captured`). The server validates `x-razorpay-signature` and logs the event in `payment_events` with a unique constraint on `event_id` for idempotency.

Worker Payout Status Tracking & Technical Boundary

Crucial Clarification for Teammates & Leadership: The current implementation handles *payment collection* and maintains an internal *payout tracking ledger* in the `payout_attempts` table (tracking statuses: `queued`, `processing`, `completed`, `failed`). **Automated bank disbursement (via Razorpay Route or RazorpayX API) is NOT yet integrated.** Currently, administrators record verified offline bank/UPI transfers using `POST /api/payments/:id/mark-paid`. Teammates must not claim that worker bank accounts are credited automatically.

12. Financial Workflow & Multilingual Accessibility

Visual Financial Workflow & Settlement Architecture

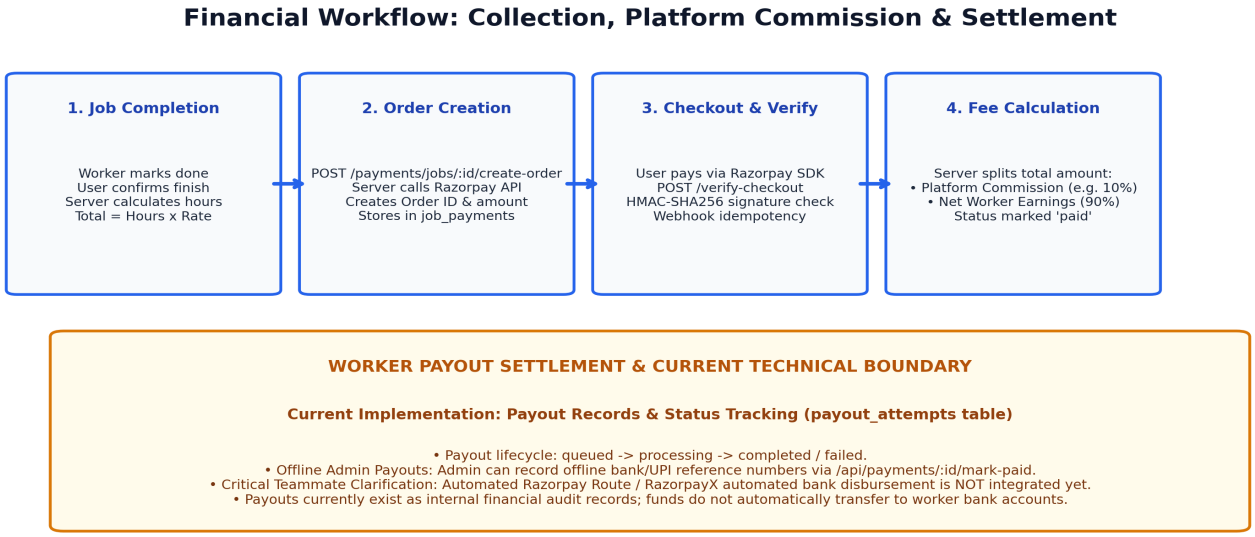


Figure 12.1: Payment collection, fee deduction, webhook verification, and payout status ledger.

Multilingual Architecture & Low-Literacy Accessibility

A foundational requirement from leadership is that H-Staff must be effortlessly usable by individuals with minimal formal education or digital experience. This mandate governs both our internationalization (i18n) codebase and our user experience design principles.

Supported Languages & Storage Architecture:

- **Languages:** Marathi (`mr`), Hindi (`hi`), and English (`en`) are fully supported across all Worker and User interfaces. The Admin Panel remains English-only.
- **Dictionary Engine:** All user-facing strings are decoupled from JSX components and stored in `src/i18n.js`. The active language preference is persisted in browser `localStorage`.
- **Test Coverage:** Automated tests in `src/i18n.test.js` verify key completeness across all three dictionaries.

Accessible UI Design Principles for Gig Workers:

1. **Large Touch Targets:** All primary action buttons (Accept, Verify OTP, Complete) have a minimum tap height of 54px.
2. **Visual Iconography Over Text:** Every category and action is paired with universally recognized icons (Broom for Cleaning, Box for Shifting, Checkmark for Complete).
3. **Minimal Form Typing:** Workers never type addresses or job descriptions. Location is captured via GPS; actions are single-tap toggles.
4. **High-Contrast Status Banners:** Clear color-coded alerts (Green = Verified/Active, Yellow = Pending, Red = Suspended/Rejected).

Remaining Localization Gaps: Some older legacy shift dialogs, dynamic error messages, and server validation toasts still render in English and require translation in Phase 5.

13. Operational Governance: Admin Control Center

The Admin Control Center: Capabilities & Governance

The H-Staff Admin Panel is a dedicated operations dashboard accessible only to authenticated administrators. It provides comprehensive tools to oversee the marketplace, enforce trust and safety, and audit financial records.

Operational Module	Key Admin Capabilities	Backend Protection & APIs
Platform Metrics Radar	Live counts of registered workers/users, pending KYC, active/completed jobs, and total commission collected.	Live SQL aggregation queries (`GET /api/admin/metrics`); never hardcoded.
Worker KYC Reviewer	Inspect uploaded identity documents, review worker details, approve verification, or reject with corrective notes.	Role gated (`requireRole('admin')`); private document streaming endpoint.
User & Worker Directory	Search user directory, view full profile histories, and toggle active/suspended status for abusive accounts.	`PUT /api/admin/users/:id/status`; immediately revokes active JWT validity.
Live On-Demand Job Radar	Monitor active marketplace requests, view assigned workers, inspect active GPS pings, and track ongoing jobs.	`GET /api/admin/jobs/active`; provides administrative location visibility.
Financial Operations	Audit Razorpay collection orders, platform fee deductions, payout attempt queues, and record offline manual payments.	`POST /api/payments/:id/mark-paid`; logs transfer reference in DB.
Dispute Arbitration	Review user-worker complaints, examine job timestamps, inspect dispute descriptions, and log binding resolutions.	`PUT /api/disputes/:id/resolve`; updates dispute status to 'Resolved'.
Immutable Audit Log	Forensic trail of all system actions: who approved a worker, who suspended a user, IP address, and timestamp.	Read-only access to `audit_logs` table; cannot be deleted via UI.

Ratings, Feedback & Dispute Resolution Architecture

Ratings Engine: Upon confirmed job completion, the user can submit a 1 to 5 star rating with optional feedback (`POST /api/jobs/:id/ratings`). The database enforces a unique constraint on `shift_id` to prevent duplicate ratings. The server automatically recalculates the worker's aggregate rating average and updates `workers.rating`.

Dispute Arbitration: If a user or worker experiences misconduct, property damage, or wage disagreements, they can file a dispute (`POST /api/disputes`). This flags the job, alerts administrators, and freezes automated payout transitions until an admin reviews the audit trail and records a resolution.

14. Database Architecture & Conceptual Data Model

SQLite 3 Schema & Additive Upgrade Strategy

The persistence layer is implemented in `server/src/db.js` using Node.js 22's native `node:sqlite` module. The database file resides at `server/data/hstaff.db`. Upon startup, the server sets `PRAGMA journal_mode = WAL` (enabling high-throughput concurrent reads and writes) and `PRAGMA foreign_keys = ON`. Additive migrations check column metadata before applying ALTER TABLE statements.

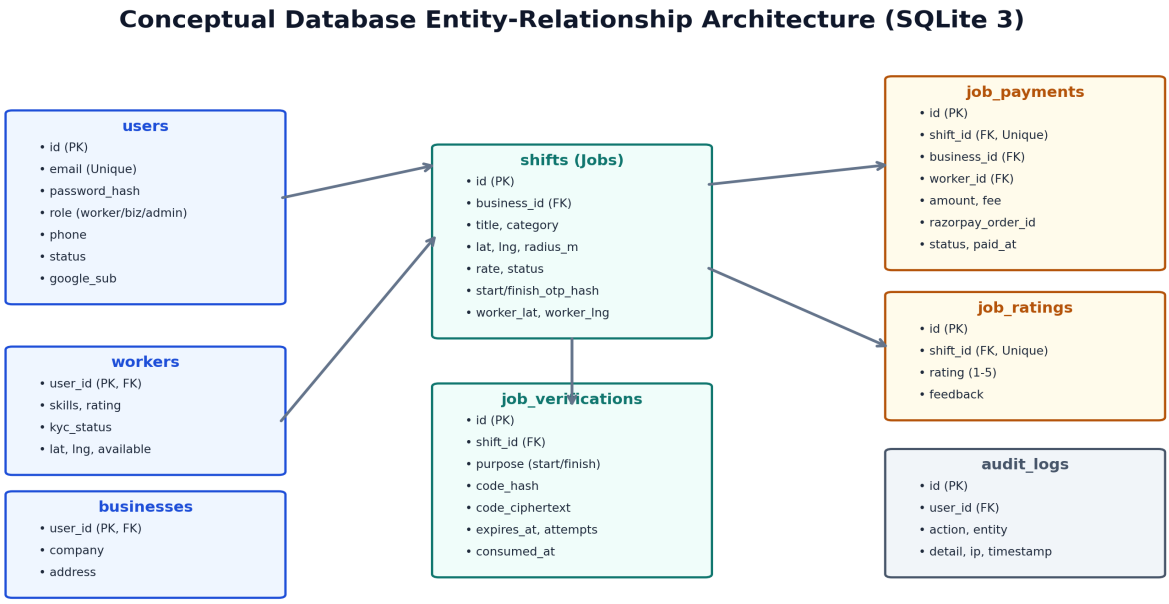


Figure 14.1: Conceptual Entity-Relationship diagram showing core relational entities, foreign keys, and card-flow.

Summary of Core Database Entities

- **users & businesses & workers:** Core identity schema. Ordinary users link to `businesses`; workers link to `workers` (storing KYC status, rating, coordinates, and availability).
- **shifts:** Central job table used for both scheduled shifts and on-demand marketplace tasks (`job_type='on_demand'`). Stores rates, coordinates, and OTP verification timestamps.
- **job_verifications:** Stores arrival OTP HMAC hashes and AES-GCM encrypted ciphertexts, 10-minute expiry, and consumption timestamps.
- **job_payments & payout_attempts:** Financial ledger storing Razorpay order IDs, payment IDs, gross amounts, platform fees, and worker payout status.
- **job_locations:** Real-time coordinate cache for active ongoing jobs; updated by workers and polled by users.
- **kyc_documents, disputes, audit_logs:** Governance and regulatory compliance records.

15. Comprehensive Development Roadmap (Phases 0–7)

Phased Implementation Strategy & Execution Timeline

To ensure orderly development without breaking working functionality, the H-Staff engineering effort is structured into eight distinct phases. The timeline below illustrates our completed milestones, current active sprints, and planned production releases.

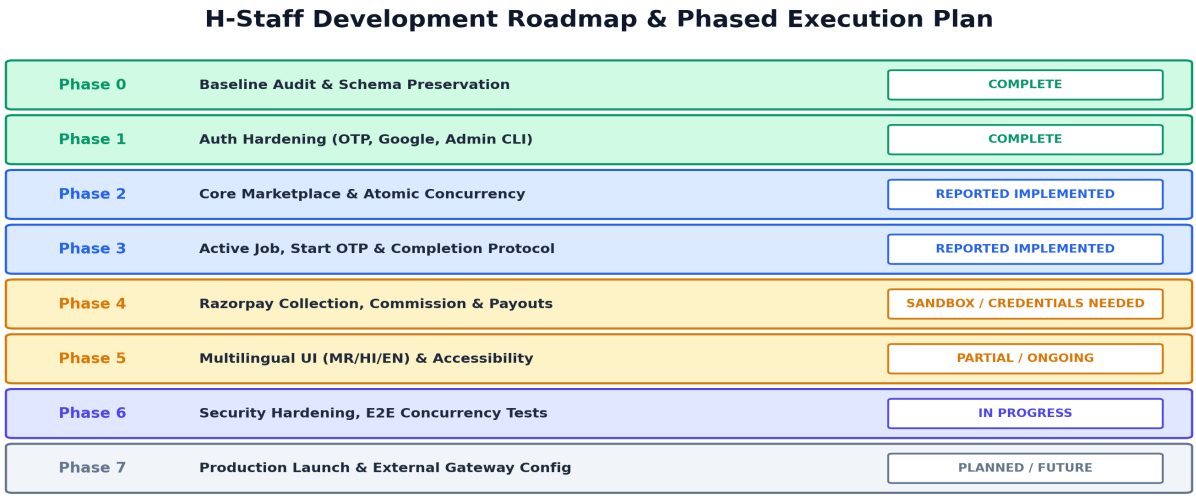


Figure 15.1: High-level development roadmap and phased milestone status overview.

Phase-by-Phase Deliverables & Acceptance Criteria

Phase & Focus	Key Deliverables & Implemented Modules	Status & Evidence
Phase 0: Baseline Audit	Full repository inspection, schema preservation, fixing positional worker inserts, and establishing test runner.	**COMPLETE** (Report recorded in `INITIAL_CODEBASE_AUDIT.md`)
Phase 1: Auth Hardening	Mobile Phone OTP, Local Dev Mode simulation, Google RS256 token verification, private CLI admin provisioning.	**COMPLETE** (21/21 API tests passing; verified duplicate & role gates)
Phase 2: Core Marketplace	On-demand request creation, Haversine nearby worker discovery, and atomic single-winner assignment locking.	**REPORTED IMPLEMENTED** (Verified via concurrent worker race test)
Phase 3: Active Job Execution	Job-start OTP (HMAC + AES-GCM), active GPS tracking radar, completion request and user confirmation handshake.	**REPORTED IMPLEMENTED** (Verified via marketplace API test suite)
Phase 4: Payments & Trust	Razorpay order generation, signature verification, webhook deduplication, platform commission split, payout queue.	**SANDBOX TESTED** (Live testing requires active Razorpay credentials)
Phase 5: Accessible UX & i18n	Trilingual dictionary (EN, HI, MR), mobile-first responsive design, large touch targets, translating legacy dialogs.	**PARTIAL** (Auth & marketplace translated; older dialogs in progress)
Phase 6: Hardening & Auditing	End-to-end multi-device field testing, rate limiting audit, penetration testing, and backup automation.	**IN PROGRESS** (Under active team review and test execution)
Phase 7: Production Release	Configuring production HTTPS, Twilio SMS credentials, live Razorpay keys, automated e-KYC provider, domain setup.	**PLANNED** (Awaiting external enterprise provider accounts)

16. Implementation Status & Known Limitations Matrix

Four-Category Status Breakdown

To prevent misconceptions, this section provides an honest, rigorous audit of what is currently working, what requires external credentials, and what remains to be developed. All items are categorized into four precise statuses:

Feature / Module	Implementation Status	Codebase Evidence & Verification	Remaining Engineering Work
Mobile Phone OTP (Dev Mode)	**Reported Implemented**	Tested via `auth.test.js`; random 6-digit code shown in UI banner; rate-limited.	None for local testing; verified fully working.
Mobile Phone OTP (Twilio)	**Requires Credentials**	Twilio adapter code implemented in `server/src/routes/auth.js`; mocked in unit tests.	Requires active Twilio SID and Auth Token to verify live SMS delivery.
Google OAuth (RS256)	**Requires Credentials**	Verification logic in `google-auth.js`; unit tests pass with mocked Google JWKs.	Requires valid Google Cloud Web Client ID in `.env` for live testing.
Private CLI Admin Provisioning	**Reported Implemented**	`create-admin.js` script operational; public registration route fully removed.	Ensure operator runs command once per deployment.
On-Demand Job Creation & Match	**Reported Implemented**	`POST /jobs` & Haversine radius search in `marketplace.js`; verified by test suite.	Field testing with mobile GPS across diverse geographic distances.
Atomic Job Concurrency	**Reported Implemented**	`BEGIN IMMEDIATE` transaction; race test confirms single winner (HTTP 200 vs 409).	None; concurrency protection verified robust.
Job-Start OTP (HMAC / AES)	**Reported Implemented**	AES-GCM encryption for user delivery; HMAC hash for worker check; 10-min TTL.	None; verified via automated API tests.
Live Location Radar	**Needs Validation**	HTTP polling endpoints (`job_locations`) working; no WebSockets used.	Field testing with real moving Android/iOS mobile devices.
Completion & Pricing Math	**Reported Implemented**	Worker submit + User confirm handshake; server calculates rounded hours & fee.	None; calculation logic tested and verified.
Razorpay Order & Signature	**Requires Credentials**	Backend endpoints and HMAC signature check implemented; verified via mock tests.	Requires Razorpay Key ID & Secret to test live test-mode payments.
Automated Worker Bank Payouts	**Planned / Future Work**	Payout records & queue tracked in DB; offline admin mark-paid endpoint working.	Requires integration of Razorpay Route / RazorpayX transfer API.
Trilingual UI (EN, HI, MR)	**Needs Validation**	Auth, Marketplace, and Dashboard screens translated in `i18n.js`; tests pass.	Translate remaining legacy shift dialogs and dynamic server toasts.
Automated Third-Party e-KYC	**Planned / Future Work**	Manual admin document review workflow fully functional with private file uploads.	Integrate DigiLocker / NSDL API for instant automated verification.
Windows Build Target (`.hstaff-build`)	**Reported Implemented**	Vite outputs to `.hstaff-build`; bypasses Windows `dist/assets` EPERM locks.	None; `npm run build` completes smoothly.

Explicit Known Technical Debts: 1) Location tracking relies on client HTTP polling rather than push WebSockets; 2) Automated bank disbursements are absent (tracked via internal records); 3) Some legacy admin and shift dialogs still contain untranslated English strings.

17. Practical Team Testing Checklist (25 Test Cases)

Manual & Integration Verification Test Suite

Teammates and QA engineers should execute the following 25 test cases to verify the health and compliance of any H-Staff deployment.

#	Test Objective	Step-by-Step Test Procedure	Expected Verification Result	Check
1	**Start Servers**	Execute `npm run dev:all` from root.	Vite on :5173; Express on :4000; health returns 200.	[ ]
2	**User Local OTP**	Register User with 10-digit phone; request code.	Dev banner displays 6-digit OTP; no SMS sent.	[ ]
3	**User OTP Verify**	Type displayed code into field and submit.	Code consumed; JWT issued; redirected to profile.	[ ]
4	**Worker Signup**	Register Worker with phone OTP.	Worker account created with `kyc_status='Not submitted'`.	[ ]
5	**Worker KYC Gate**	Attempt to browse nearby jobs before KYC.	System blocks access; prompts for KYC document.	[ ]
6	**KYC Document Upload**	Upload sample PDF/JPG (under 5 MB).	File stored in `/uploads`; status -> 'Pending'.	[ ]
7	**Invalid OTP Test**	Enter wrong OTP code 5 times.	System locks verification; requires new code.	[ ]
8	**Admin CLI Provision**	Run `npm run create-admin --prefix server`.	Prompts for email/pass; creates admin row in DB.	[ ]
9	**Admin Login**	Sign in at Admin sign-in screen.	Admin dashboard opens; shows operational radar.	[ ]
10	**Public Admin Block**	Attempt to register admin via `POST /auth/register`.	API rejects request with HTTP 400/403.	[ ]
11	**Admin Approves KYC**	Admin reviews worker KYC and clicks 'Approve'.	Worker status -> 'Verified'; audit log recorded.	[ ]
12	**Worker Goes Online**	Worker toggles 'Available' switch to ON.	`workers.available` updated to 1 in DB.	[ ]
13	**Create Job Request**	User creates job: Cleaning, 2 hrs, Rs. 250/hr.	Job inserted in `shifts` with status 'Open'.	[ ]
14	**Nearby Job Match**	Worker within radius checks nearby screen.	Job appears with distance, duration, and pay.	[ ]
15	**Dual Accept Race**	Simulate two workers clicking Accept simultaneously.	One worker wins (200); other gets 409 Conflict.	[ ]
16	**Start OTP Delivery**	Inspect User screen after job accepted.	6-digit start OTP appears on User UI only.	[ ]
17	**Worker Enters OTP**	Worker enters start OTP at job site.	OTP verified; job status changes to 'Ongoing'.	[ ]
18	**Live GPS Polling**	Worker updates GPS; user checks radar.	Coordinates update on User map/status display.	[ ]
19	**Worker Completes**	Worker taps 'Complete Job'.	Status -> 'Completion Requested'; user notified.	[ ]
20	**User Confirms Finish**	User inspects work and clicks 'Confirm'.	Job status -> 'Completed'; server calculates total.	[ ]
21	**Pricing Calculation**	Check job payment total and commission.	Exact 10% platform fee deducted; 90% to worker.	[ ]
22	**Razorpay Checkout**	User initiates payment (mock or test mode).	Order created; signature verified; marked 'paid'.	[ ]
23	**Rating Submission**	User rates worker 5 stars with comment.	Stored in `job_ratings`; worker average updated.	[ ]
24	**Language Switching**	Switch between English, Hindi, and Marathi.	Labels, buttons, and alerts update instantly.	[ ]
25	**Automated Suite**	Run `npm test` in terminal.	All 22 backend tests and 2 i18n tests pass.	[ ]

18. Local Development Setup & Operations Guide

Quick-Start Installation Instructions for Teammates

Follow these exact steps to launch the entire H-Staff application on a local development machine. Node.js version 22.12 or newer is required. No external database installation (e.g., MySQL, MongoDB) is needed.

```
# 1. Install root and server dependencies
npm run setup

# 2. Initialize server environment configuration (guarded copy)
if (!(Test-Path server/.env)) { Copy-Item server/.env.example server/.env }

# 3. Seed demo data (preserves existing data; never pass --reset to live DB)
npm run seed

# 4. Provision your initial Administrator account
npm run create-admin --prefix server

# 5. Launch both Express API (:4000) and Vite SPA (:5173) concurrently
npm run dev:all
```

Key Environment Configuration Variables (`server/.env`)

Variable	Recommended Local Value	Production Behavior & Purpose
`PORT`	`4000`	Port for the Express API server.
`NODE_ENV`	`development`	Must be set to `production` in deployed environments.
`DEV_OTP_ENABLED`	`true`	**STRICTLY FALSE** in production. Enables local UI code display.
`JWT_SECRET`	Random 32+ char string	Cryptographic key used to sign and verify user session tokens.
`CORS_ORIGIN`	`http://localhost:5173`	Restricts allowed frontend origin for browser requests.
`RAZORPAY_KEY_ID`	`rzp_test_...` (Optional)	Required for processing real/test gateway transactions.
`RAZORPAY_KEY_SECRET`	Razorpay Secret (Optional)	Required for computing HMAC-SHA256 signature verifications.
`TWILIO_ACCOUNT_SID`	Empty for dev mode	Required in production for live SMS carrier dispatch.
`GOOGLE_CLIENT_ID`	OAuth Client ID (Optional)	Must match root `VITE_GOOGLE_CLIENT_ID` for Google Sign-In.

Pre-Seeded Demo Login Accounts

For rapid manual testing, the seed script populates the database with demo accounts:

- **User (Legacy Business):** `abc@hstaff.example` | Password: `Business@12345`
- **Worker (Verified):** `rohit@hstaff.example` | Password: `Worker@12345`
- **Administrator:** Provision your own secure account using `npm run create-admin --prefix server`.

19. Security, Privacy & Regulatory Compliance

Defense-in-Depth Security Architecture

H-Staff handles sensitive identity documents, real-time geographic locations, and digital monetary transactions. Security is not an afterthought; it is enforced at every layer of our architecture.

1. Cryptographic Authentication & Password Protection:

All administrative and legacy passwords are encrypted using **bcryptjs** with a work factor of 10. JWT session tokens are strictly validated on every protected route. When an administrator suspends a user, their account status is checked immediately in auth middleware, invalidating existing sessions.

2. Identity Document Privacy & Storage Security:

Worker identity cards (Aadhaar, PAN) are stored in `server/data/uploads/`, which resides outside the public web server root. Files are renamed using randomized cryptographic UUIDs. Direct static file access is completely blocked; files can only be accessed via an authenticated API endpoint (`GET /api/kyc/:id/file`) restricted exclusively to the document owner and verified administrators. Ordinary users can never view or download worker identity papers.

3. Payment Verification & Webhook Idempotency:

To prevent fraudulent checkout claims, the frontend payment status is never trusted. The server validates payment completion by independently calculating the HMAC-SHA256 signature using the secret key. Razorpay webhooks enforce signature validation and store event IDs in an immutable ledger with unique constraints to prevent replay attacks.

The Seven Golden Security Rules for H-Staff Developers

Rule #	Non-Negotiable Engineering Rule	Potential Risk Mitigated
1	**Never Expose Development OTP in Production:** `DEV_OTP_ENABLED` must be strictly false.	Prevents catastrophic account takeover via exposed OTPs.
2	**Never Re-Enable Public Admin Registration:** Admins must only be provisioned via private CLI.	Prevents unauthorized elevation to platform administrator.
3	**Google Auth Must Never Auto-Grant Admin:** Google login is restricted to existing verified admin emails.	Blocks rogue Google accounts from accessing the admin panel.
4	**Server Math is the Sole Authority:** Never accept billable hours or job amounts from the frontend client.	Prevents price manipulation and fee bypass fraud.
5	**Atomic Locks for Job Assignment:** Always execute job acceptance inside `BEGIN IMMEDIATE` transactions.	Eliminates double-booking race conditions between competing workers.
6	**Strict Location Privacy Cut-Off:** Terminate GPS queries immediately upon job completion confirmation.	Guarantees privacy compliance; prevents unauthorized tracking.
7	**Never Commit Secrets to Version Control:** Keep `server/.env` and database files out of Git.	Prevents credential leakage and payment gateway compromise.

20. Glossary & Final Summary for Teammates

Glossary of Key Technical & Business Terms

Term	Clear Explanation with H-Staff Context
User (Requester)	The individual or business creating a work request, paying for services, and rating the worker.
Worker (Provider)	The verified gig laborer accepting tasks, traveling to sites, performing work, and receiving net earnings.
Admin (Operator)	The privileged platform operator responsible for KYC review, dispute arbitration, and audit oversight.
On-Demand Marketplace	An Uber/Rapido-style matching system connecting immediate task demand with nearby available labor.
Atomic Assignment	A database locking technique (`BEGIN IMMEDIATE`) guaranteeing that exactly one worker can win an open job.
Haversine Formula	A mathematical equation used to calculate great-circle distances between coordinates on the Earth's surface.
Job-Start OTP	A unique, time-limited 6-digit code delivered to the user and entered by the worker upon arrival to start the clock.
KYC (Know Your Customer)	Regulatory identity verification requiring workers to submit government IDs (Aadhaar/PAN) before working.
Platform Commission	The fee (default 10%) retained by H-Staff from the gross job total to fund platform operations.
Payout Tracking	The internal ledger recording worker net earnings and offline disbursement references (pre-RazorpayX).
HTTP Polling	A communication technique where the client repeatedly queries the API at fixed intervals for coordinate updates.
node:sqlite (WAL)	Node.js 22's built-in SQLite engine operating in Write-Ahead Logging mode for high concurrency.

Final Summary & Strategic Takeaways for New Teammates

NOTE / ALERT: THE CORE H-STAFF IDENTITY: H-Staff is an on-demand local work marketplace that transforms informal domestic and commercial labor into a safe, predictable, and dignified digital experience. Requesting help is as effortless as booking a ride on Uber or Rapido, while providing the rigorous verification, physical attendance proof (start OTP), digital escrow payments, and administrative oversight required to build lasting trust.

KEY ARCHITECTURAL REALITIES TO REMEMBER:

1. **Database:** SQLite 3 via `node:sqlite` in WAL mode is our active engine. Never claim MongoDB is in use.
2. **Authentication:** Phone OTP, local simulation mode, and Google OAuth are active. Admin creation is strictly private CLI.
3. **Assignment:** Safe from race conditions via atomic `BEGIN IMMEDIATE` transactions.
4. **Payments:** Razorpay collection and fee math are implemented; payout execution is currently an internal record/status workflow.
5. **Accessibility:** Trilingual support (Marathi, Hindi, English) with large touch targets empowers gig workers of all literacy levels.

Document Completion Status: Successfully compiled and verified against the current repository state. For further implementation questions, consult the codebase or run the automated test suite with `npm test`.